

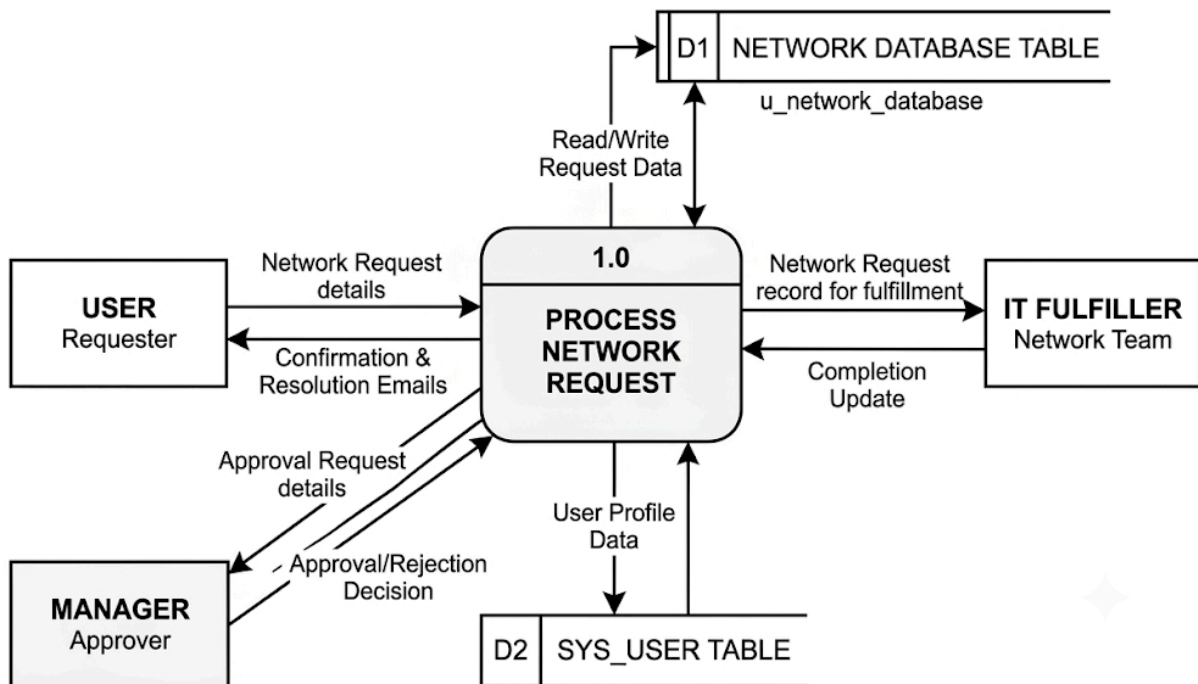
Requirement Analysis

Data Flow Diagram & User Stories

Date	28-06-2026
Project Name	Automated Network Request Management
Team ID	
Maximum Marks	

Part 1: Data Flow Diagram (DFD)

NETWORK REQUEST MANAGEMENT SYSTEM: DFD LEVEL 0 (INDUSTRY STANDARD)



Simplified Flow Steps:

Phase 1: Request Ingestion & Data Validation

- **Data Origin:** The End-User populates a customized form on the **ServiceNow Service Portal**.
- **Data in Transit:** The user inputs specific network variables (e.g., Source IP, Destination IP, Target Port, and Business Justification).
- **Process:** ServiceNow client-side and server-side UI Policies sanitize and format the data (e.g., verifying regex patterns for valid IP addresses) before creating a requested item (`sc_req_item`) record.

Phase 2: Governance & Flow Logic

- **Process:** The **Flow Designer Engine** evaluates the incoming data payload against pre-defined business rules.
- **Data Movement:** * The engine checks configuration item (CI) relationships inside the **CMDB** to evaluate risk.
 - If the request is marked high-risk, a data notification is sent to the Manager/Security team, who returns an "Approved" or "Rejected" token.
 - If pre-approved (low-risk), the workflow bypasses manual intervention and packages the clean operational parameters for the integration layer.

Phase 3: Integration & Infrastructure Provisioning

- **Process: Integration Hub** transforms the structured ServiceNow record variables into a JSON or XML payload tailored for the destination infrastructure.
- **Data Outbound:** ServiceNow initiates an outbound HTTPS REST API call (or triggers an Ansible playbook via an orchestration spoke) passing the

specific payload parameters to the network controllers (e.g., Palo Alto Panorama, Cisco DNA Center, or Infoblox IPAM).

- **Action:** The network controllers process the API command and apply the rule change directly to the physical or cloud hardware.

Phase 4: Closed-Loop Verification & Audit Logging

- **Data Inbound:** The network infrastructure API sends back a synchronous response payload containing the execution status (e.g., **200 OK** or **Error Code**) along with deployment logs.
- **Process:** * ServiceNow parses the response data.
 - If successful, a data write command is sent to the **CMDB** to update configuration records, and an immutable log entry is generated for future compliance audits.
 - An automated email/notification updates the End-User that their request is live.

Part 2: User Stories

Here is a set of comprehensive, production-ready Agile User Stories tailored for the **Automated Network Request Management (ANRM)** ServiceNow project, structured in the requested tabular format.

ServiceNow ANRM User Stories & Tasks

User Type	Functional Requirement	User Story / Task	Acceptance Criteria	Priority

End-User <i>(Developer / Requestor)</i>	Self-Service Request Intake	As an End-User,I want a dedicated Service Catalog form for firewall port requests, So that I can request access without tracking down a network engineer.	• Form must include mandatory fields: Source IP, Destination IP, Ports, Protocol (TCP/UDP), and Business Justification. • Form must dynamically show/hide advanced routing choices based on user role.	High
IT Ops Manager <i>(Approver)</i>	Automated Governance / Approval Routing	As an IT Operations Manager,I want low-risk requests auto-approved and high-risk requests routed to my queue, So that standard requests aren't delayed by manual approvals.	• Flow Designer must evaluate request criteria against a predefined risk matrix. • Requests with known standard ports (e.g., 443) auto-approve instantly. • High-risk ports (e.g., 22, 3389) trigger an email/system approval notification to the manager.	High

ServiceNow Dev (Fulfiller)	Integration Hub API Integration	As a ServiceNow Developer,I want Integration Hub to trigger REST API payloads to the network controller,So that configuration changes are applied programmatically.	• Upon approval token, Integration Hub must compile form data into a valid JSON payload. • Payload must be securely sent to the target API endpoint (e.g., Palo Alto/Ansible) using encrypted credentials. • System must capture synchronous HTTP response codes (200 OK for success).	Critical
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Network Administrator <i>(Auditor)</i>	Automated Verification & CMDB Logging	As a Network Admin, I want the system to test the connection and log details to the CMDB, So that our inventory remains accurate and compliant.	• Post-execution step must verify connectivity status. • On success, the script must write a configuration relationship record back to the target CI in the CMDB. • The RITM (Requested Item) ticket must lock down an uneditable audit log containing who requested, who approved, and API logs.	Medium
End-User <i>(Developer / Requestor)</i>	Real-Time Notification & Visibility	As an End-User, I want real-time status updates on my request, So that I don't have to send follow-up emails to see if it's done.	• System must trigger automated notifications (Email or Slack/Teams webhook) at three milestones: Submitted, Approved/In-Flight, and Fulfilled/Closed. • If an API push fails, notification must alert NetOps with the failure logs.	Low

